

Power System Influences

Energy Transmission
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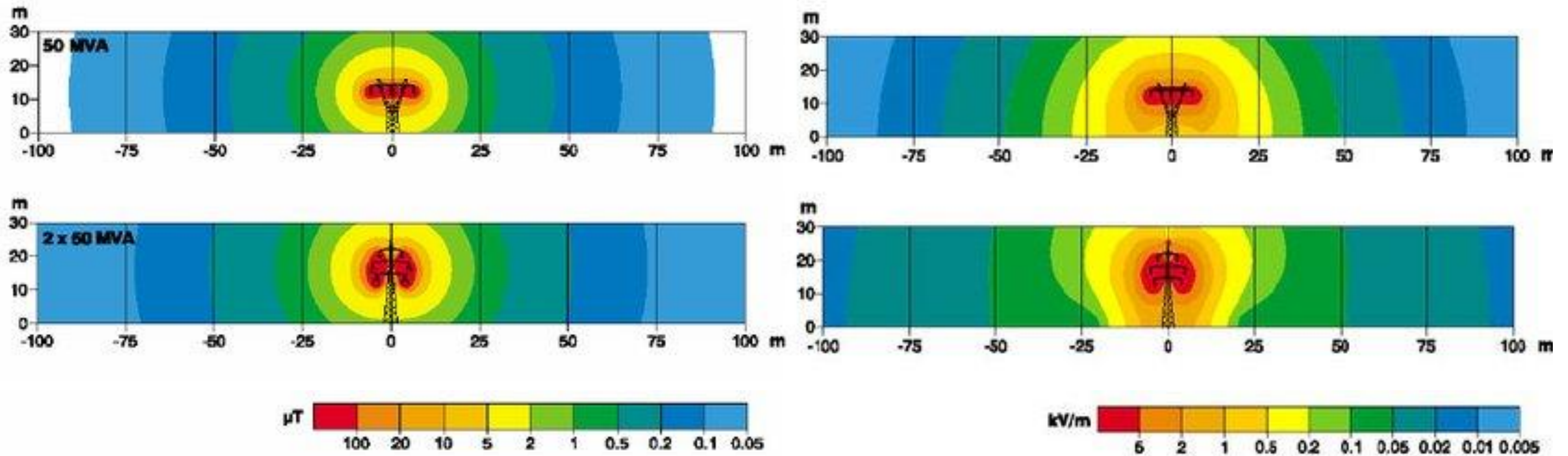
- ♦ Capacitive influence
- ♦ Inductive influence
- ♦ Exercise:
 - ♦ Capacitive coupling matrix
 - ♦ Railway transmission line and pasture fence

Influence of power lines

- ◆ Transmission lines are sources of electromagnetic emissions
 - leaving the station facilities; they may thus interact with virtually every other line system
 - Routes are bundled; thus long parallel segments with strong capacitive and inductive coupling



Fields in the vicinity of overhead lines



- ◆ Typical levels for the magnetic field (on the left) and electric field (on the right) in the vicinity of the 150kV overhead power lines used in Greece.

Electromagnetic influences

- ◆ Undesired interaction between electrical power systems and their surrounding due to
 - Magnetic fields
 - Electrical fields
- ◆ Electromagnetic compatibility (EMC)
 - Asset can withstand their electromagnetic environment without significant restrictions
 - Asset limits own emissions to an acceptable level
- ◆ Danger to human beings and livestock
 - Touch voltages (may be even in unexpected places)

Severity of touch voltages

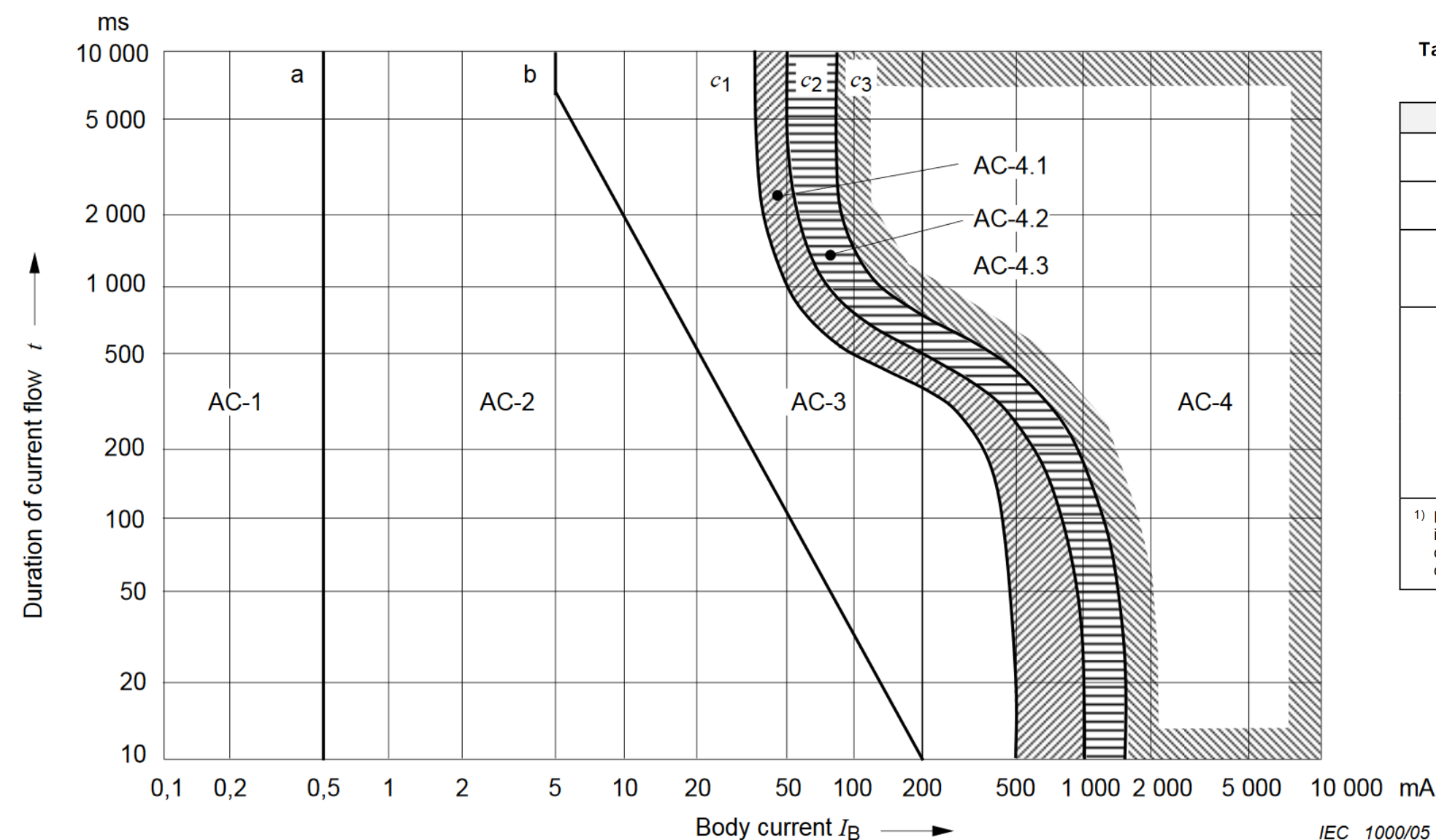


Table 11 – Time/current zones for a.c. 15 Hz to 100 Hz for hand to feet pathway – Summary of zones of Figure 20

Zones	Boundaries	Physiological effects
AC-1	Up to 0,5 mA curve a	Perception possible but usually no 'startled' reaction
AC-2	0,5 mA up to curve b	Perception and involuntary muscular contractions likely but usually no harmful electrical physiological effects
AC-3	Curve b and above	Strong involuntary muscular contractions. Difficulty in breathing. Reversible disturbances of heart function. Immobilization may occur. Effects increasing with current magnitude. Usually no organic damage to be expected
AC-4 ¹⁾	Above curve c ₁ c ₁ -c ₂ c ₂ -c ₃ Beyond curve c ₃	Patho-physiological effects may occur such as cardiac arrest, breathing arrest, and burns or other cellular damage. Probability of ventricular fibrillation increasing with current magnitude and time AC-4.1 Probability of ventricular fibrillation increasing up to about 5 % AC-4.2 Probability of ventricular fibrillation up to about 50 % AC-4.3 Probability of ventricular fibrillation above 50 %

¹⁾ For durations of current flow below 200 ms, ventricular fibrillation is only initiated within the vulnerable period if the relevant thresholds are surpassed. As regards ventricular fibrillation, this figure relates to the effects of current which flows in the path left hand to feet. For other current paths, the heart current factor has to be considered.

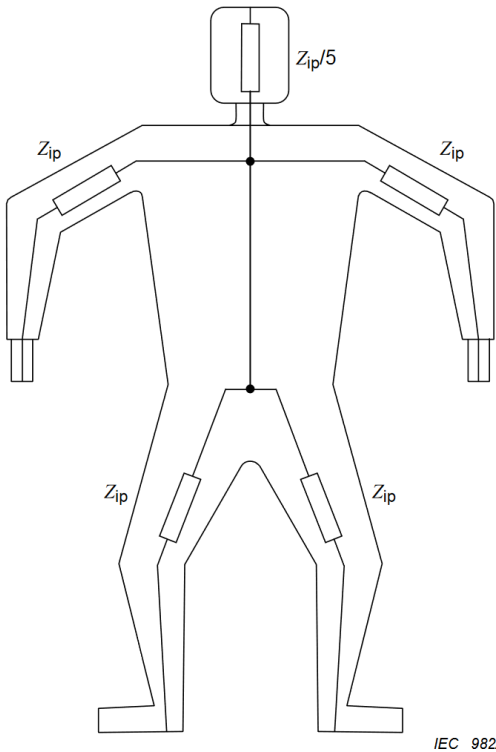


Figure 20 – Conventional time/current zones of effects of a.c. currents (15 Hz to 100 Hz) on persons for a current path corresponding to left hand to feet

Assets prone to influence from power lines



◆ Telephone lines

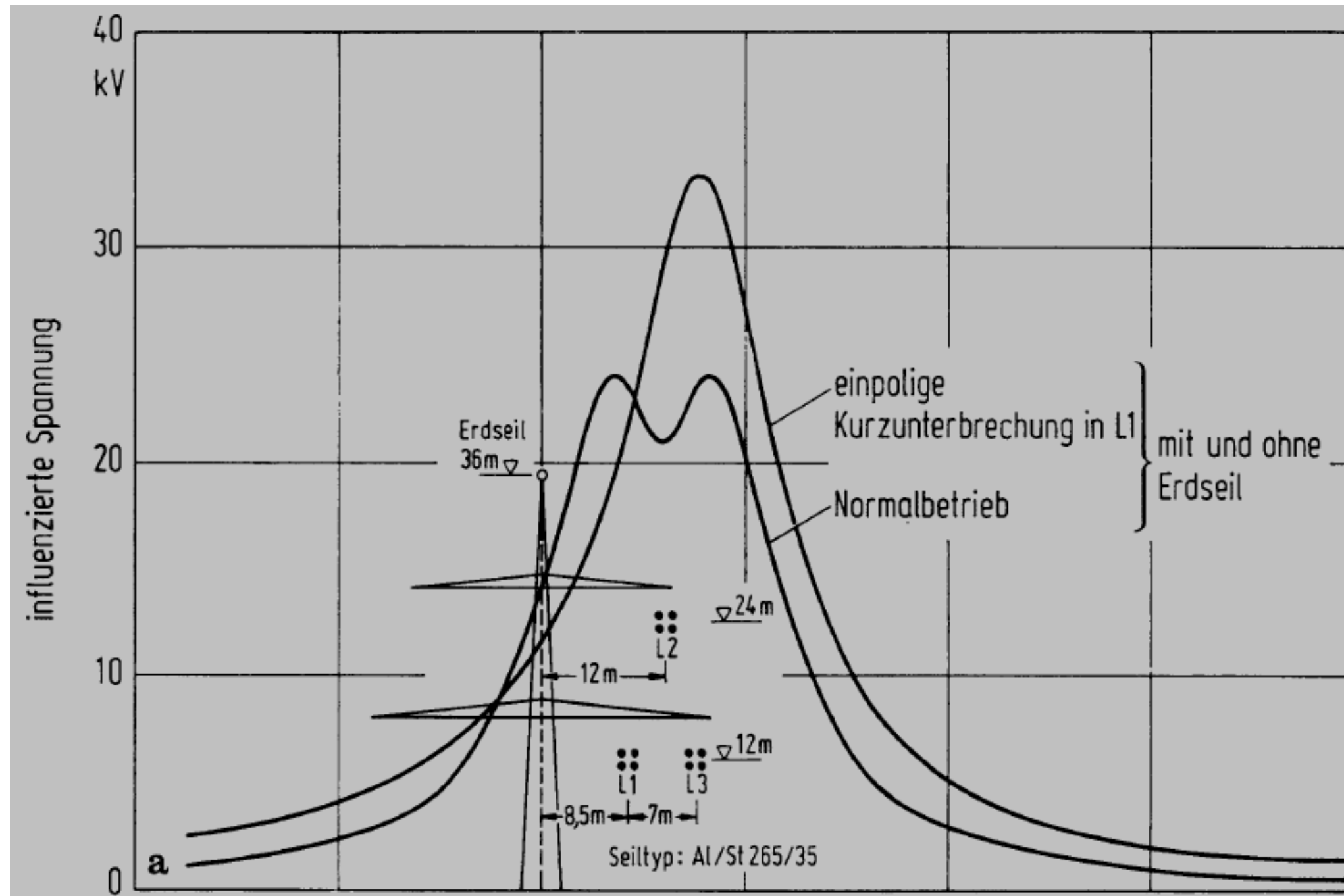


◆ Pipelines

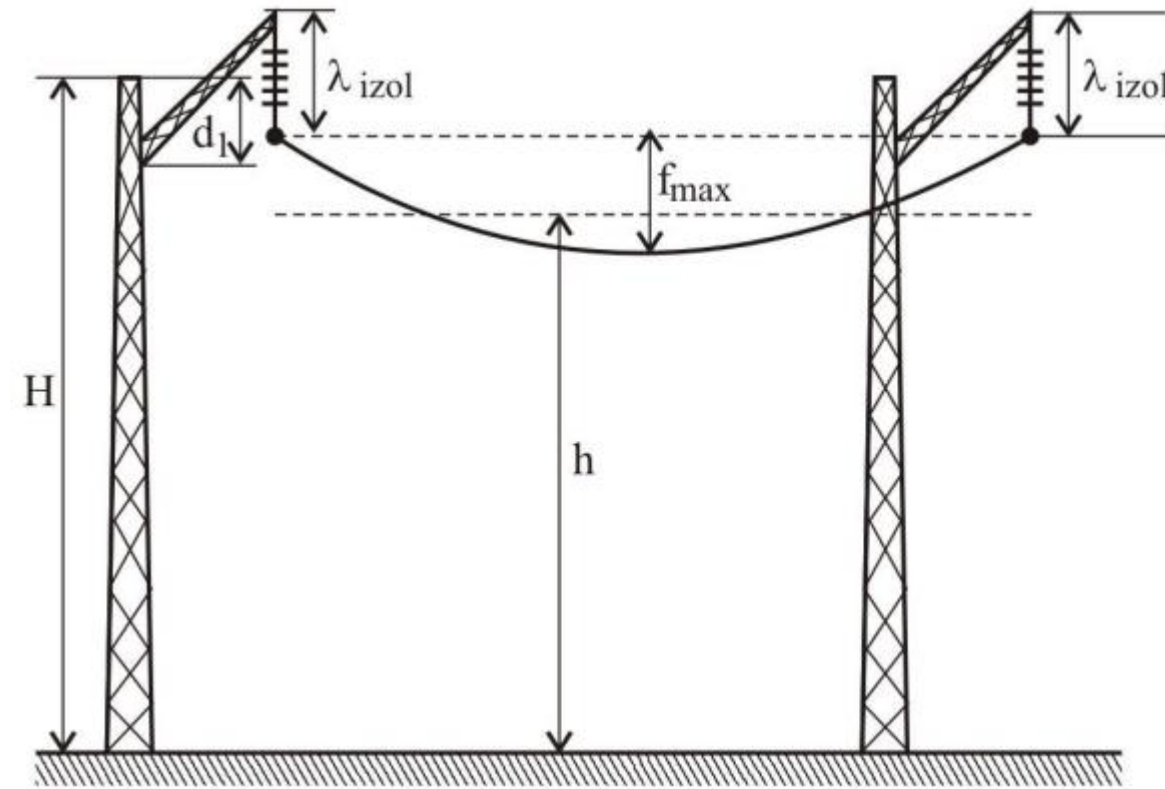
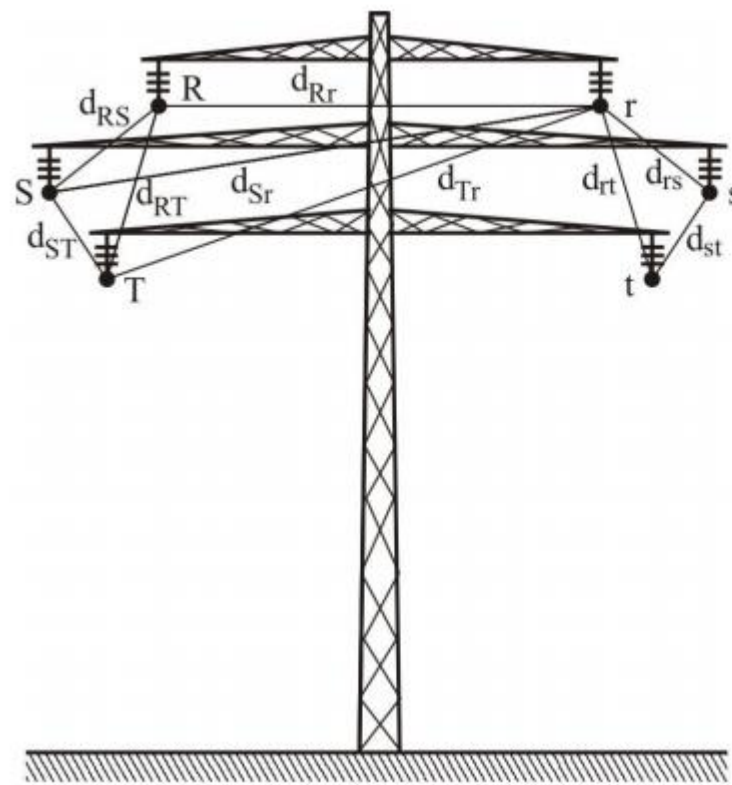


◆ Wire fences

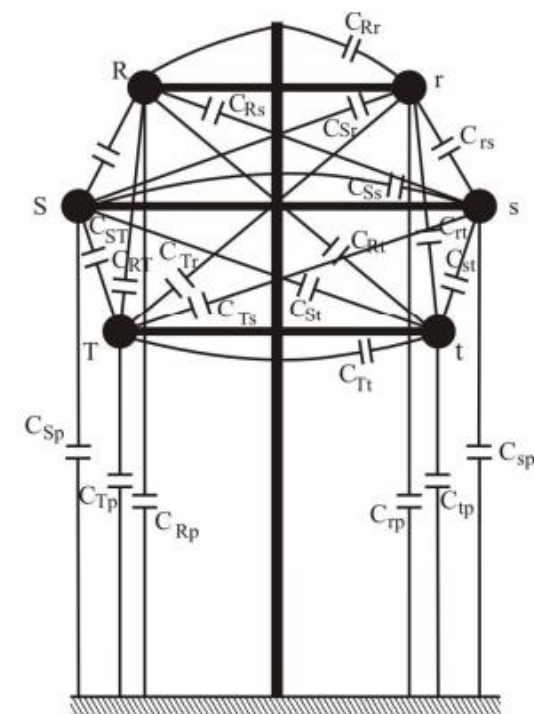
Capacitive influence on infinite wire 5 m above ground



Capacitive influence



- ◆ Determine geometric properties



- for the partial capacities between phases:

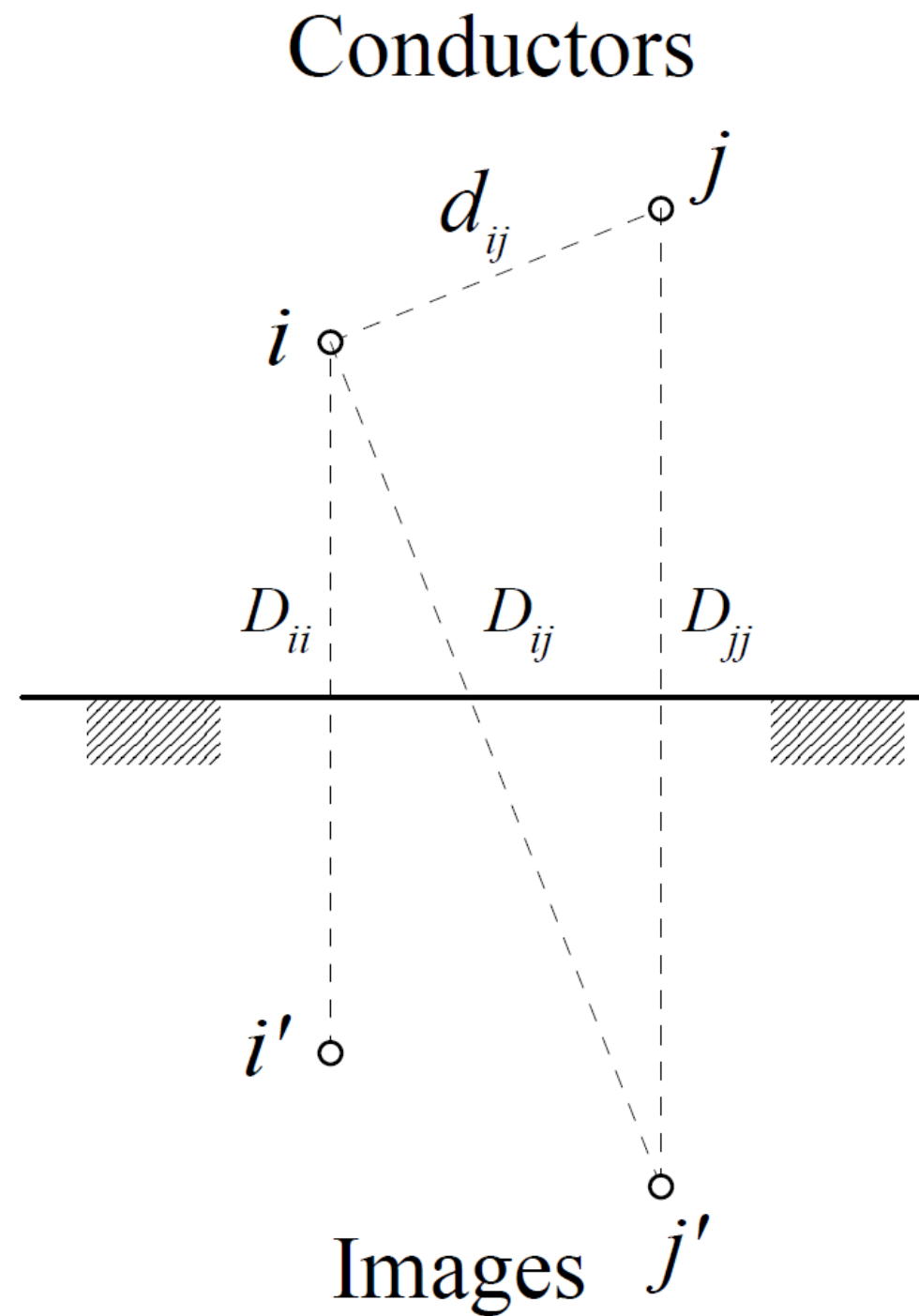
$$C_{ik} = \frac{2\pi\epsilon_0 l}{\ln \frac{d_{ik}}{r_0}} \quad (5)$$

- for the partial capacities between phases and ground:

$$C_{pi} = \frac{2\pi\epsilon_0 l}{\ln \left(\frac{2h_i}{r_0} \right)} \quad (6)$$

- ◆ Coupling capacitance matrix

Example: Capacitance matrix



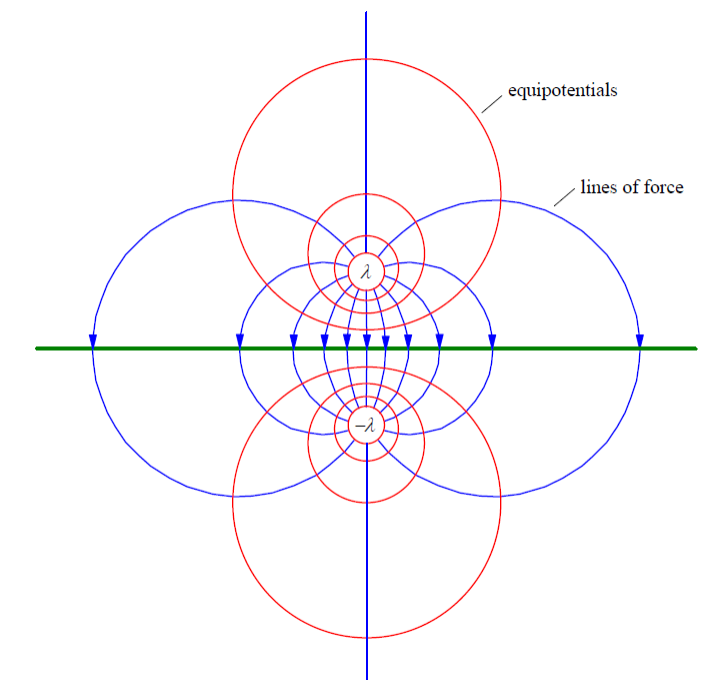
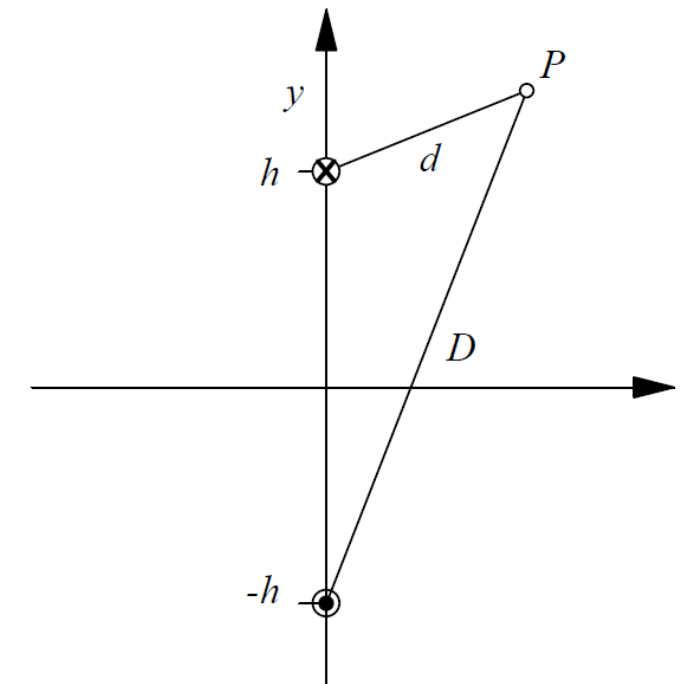
$\mathbf{A}$ = matrix with elements $A_{ij} = \ln \frac{D_{ij}}{d_{ij}}$

$d_{ii} = R_i$ = radius of conductor i

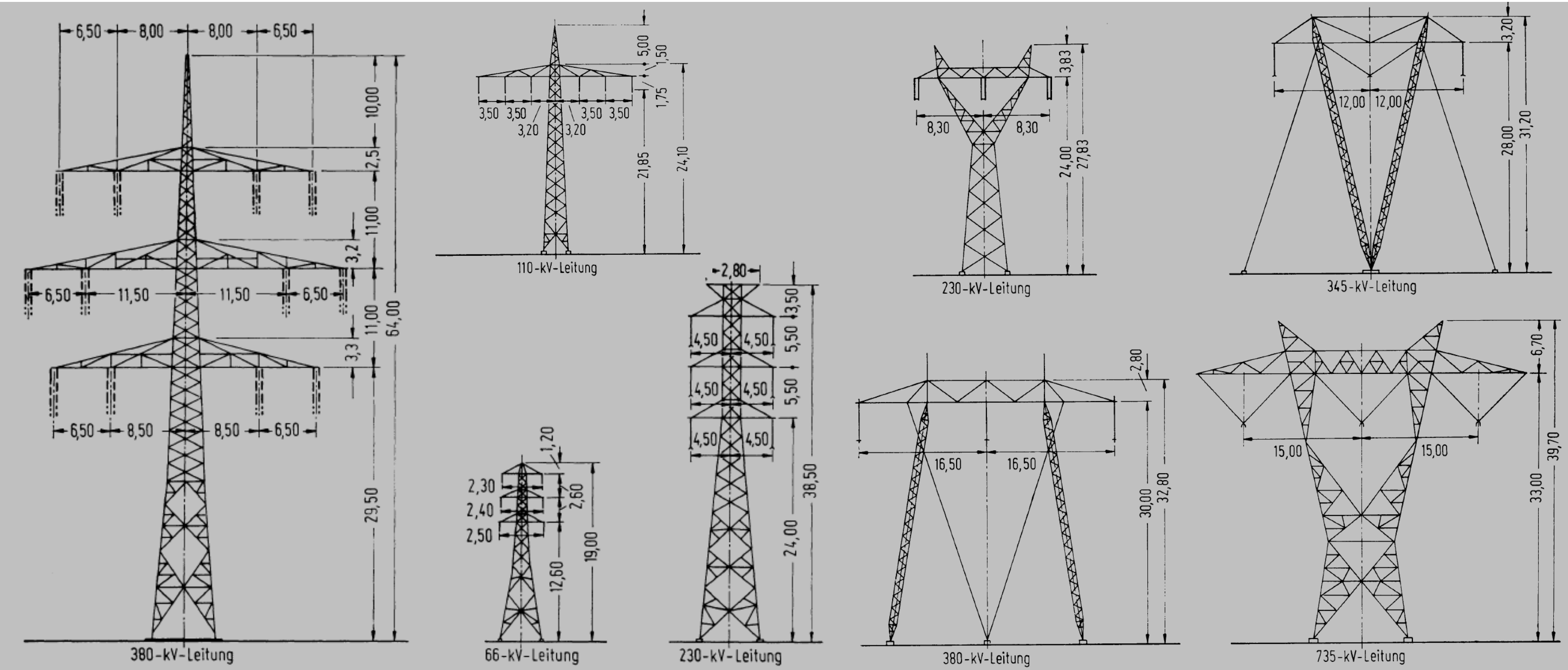
$D_{ii} = 2h_i$ = double height of conductor i

$$\mathbf{A} = \begin{bmatrix} \ln \frac{D_{ii}}{d_{ii}} & \ln \frac{D_{ij}}{d_{ij}} \\ \ln \frac{D_{ji}}{d_{ji}} & \ln \frac{D_{jj}}{d_{jj}} \end{bmatrix}$$

$$\mathbf{C} = 2\pi\epsilon_0 \mathbf{A}^{-1} \quad \text{Fm}^{-1}$$

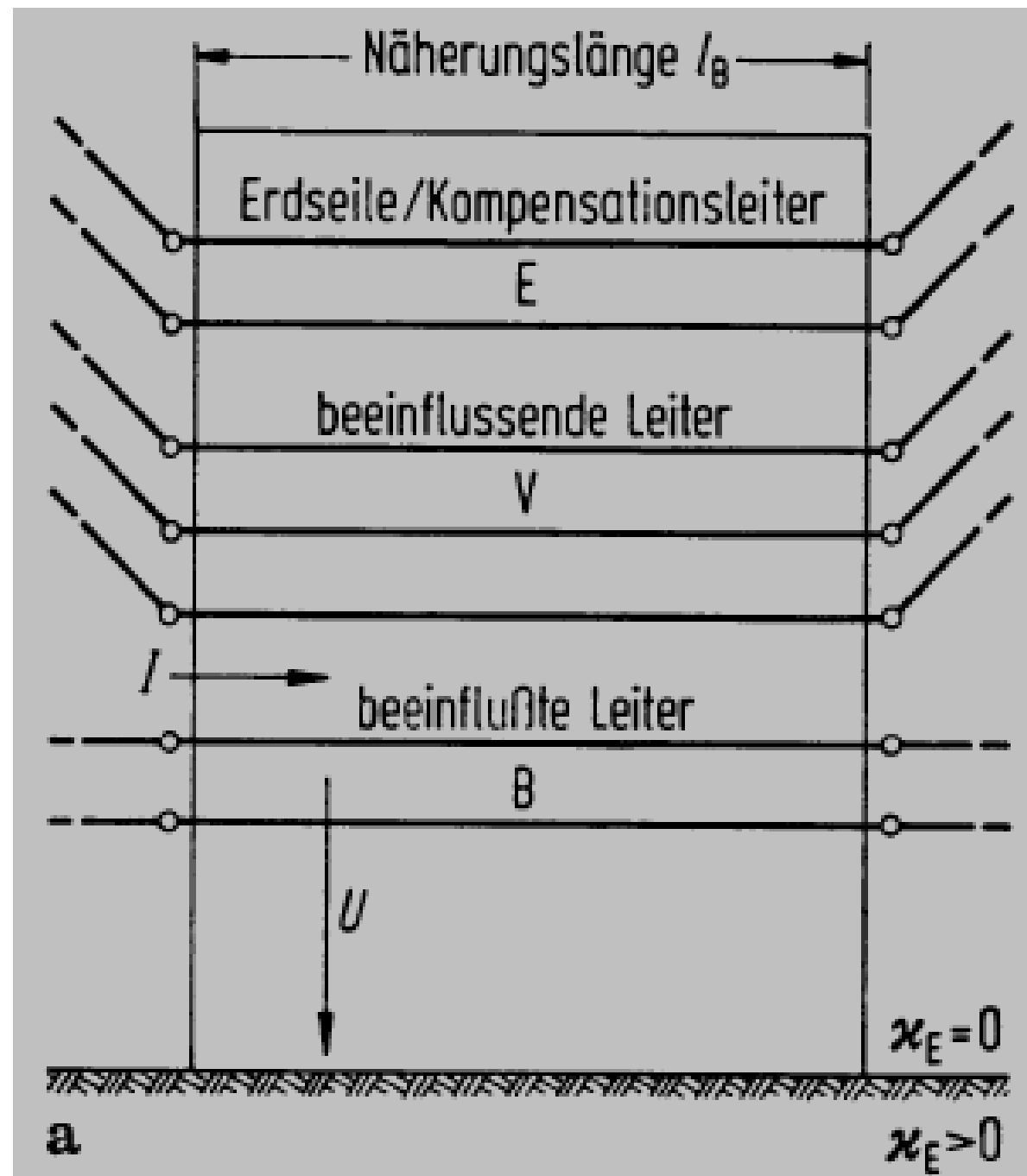


Examples of conductor distance for different tower types



(Hosemann 2001)

Calculating capacitive influence



- ◆ Wires are modeled as lines parallel to each other
- ◆ Known quantities: Voltages and currents of the influencing wires, coupling matrix
- ◆ Out of those, voltages and currents of the influenced wires are calculated
- ◆ More complex geometries are broken down in a series of approximately parallel segments

$$\begin{bmatrix} \underline{q}'_V \\ \underline{q}'_B \end{bmatrix} = \begin{bmatrix} C'_{VV} & C'^T_{BV} \\ C'_{BV} & C'_{BB} \end{bmatrix} \cdot \begin{bmatrix} \underline{u}_V \\ \underline{u}_B \end{bmatrix}$$

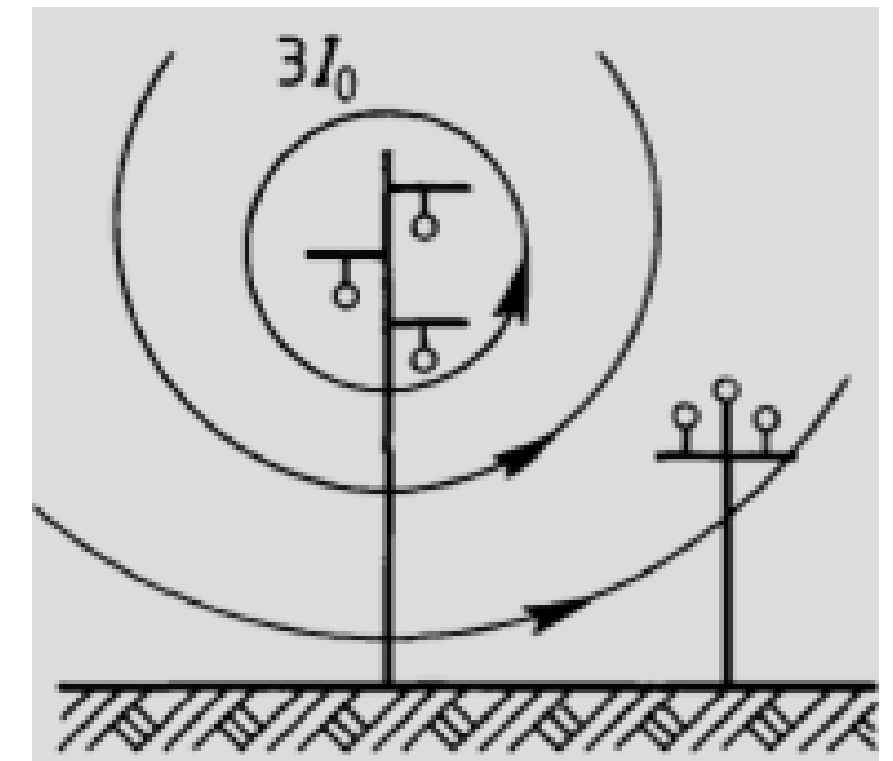
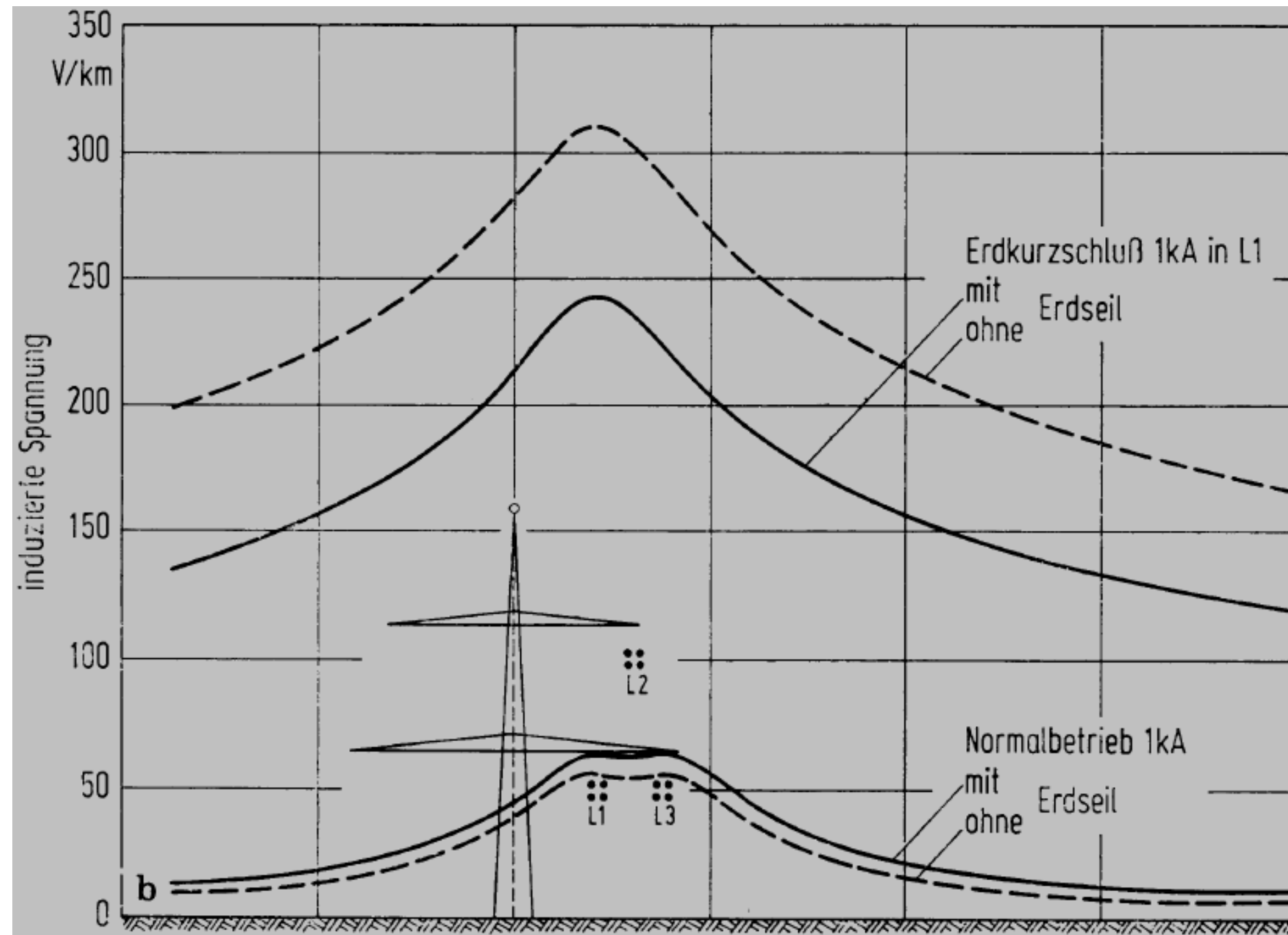
Example of capacitive coupling effect

Table 4. Comparison between the measured and calculated capacitively induced voltages

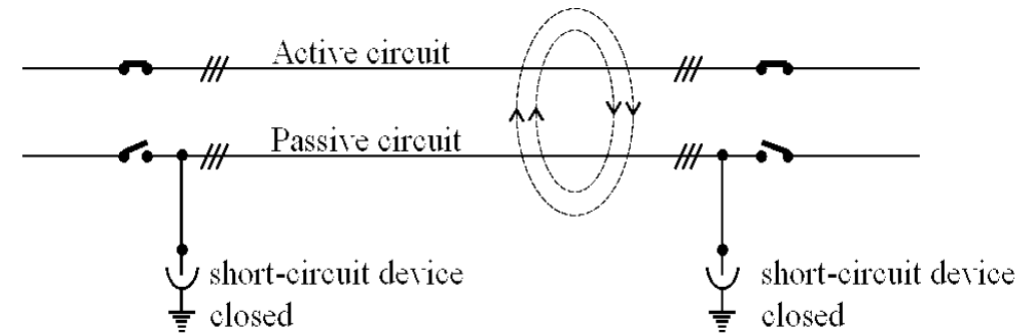
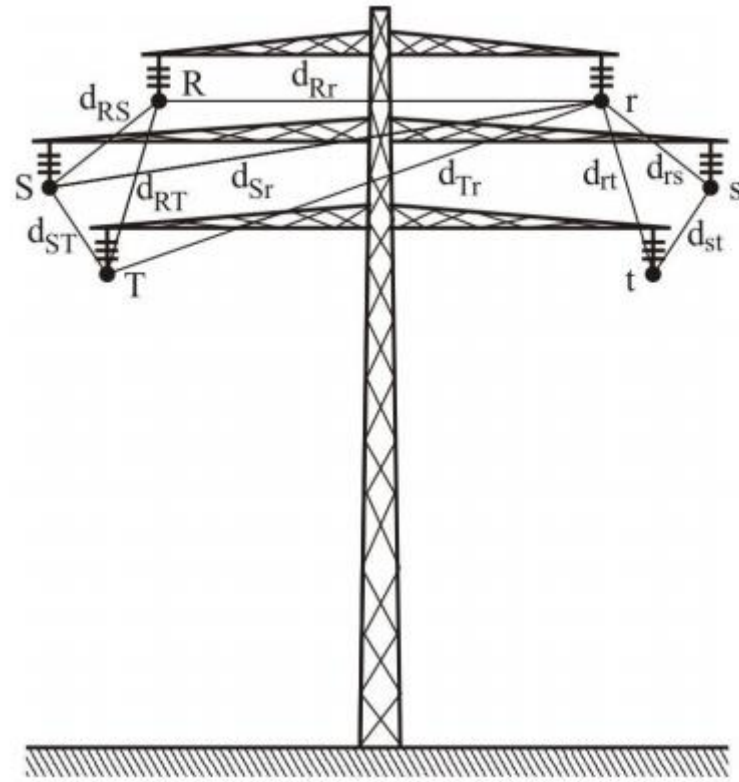
Overhead electric line	Circ. 1 [km]	Circ. 2 [km]	Activ circuit voltage	Capacitively induced voltage measured in passivated circuit			Capacitively induced voltage calculated in passivated circuit		
			U [kV]	U _R [kV]	U _S [kV]	U _T [kV]	U _r [kV]	U _s [kV]	U _t [kV]
Mintia-Hasdat	49.876	25.455	237.5	8.87	2.7	4.4	8.823	2.755	4.415
Pestis.-Otelarie max. charge	25.455	11.249	236.9	12.7	20.2	18.3	12.643	20.324	12.278
Pestis-Otelarie	25.455	11.249	236.9	12.7	20.2	12.3	12.726	20.324	12.278
Baru M-Hasdat	16.688	43.897	225	1.9	3.35	2.35	1.845	3.419	2.427
Otelarie-Hasdat	25.455	7.422	236.8	19.4	23.4	18.2	19.433	23.259	17.906
PdF-Resita	116.55	116.55	228	10.4	3.6	5.1	10.422	3.619	4.948
			225						
			232						
Mintia-Pestis	18.675	18.675	237	8.9	4.42	6.25	9.076	4.573	6.278
Resita-Iaz	30.730	30.730	226.5	3.03	7.94	5.9	3.154	7.998	5.876
Resita-Timisoara	72.867	72.867	234	11.1	3.7	5.4	11.221	3.768	4.798
Timisoara-Sacalaz	53.719	24.620	230	6.55	5.82	5.41	6.582	5.865	5.452
			235						
			225						
Sacalaz-Arad	53.719	55.173	230	8.2	2.8	4.2	8.206	2.845	4.210
			235						
			225						

(Suaranu 2009)

Inductive influence on infinite wire 5m above ground



Inductive influence



$$\begin{aligned} i_R &= \sqrt{2} \cdot I_{fR} \sin(\omega \cdot t + \varphi) \\ i_S &= \sqrt{2} \cdot I_{fS} \sin\left(\omega \cdot t + \varphi - \frac{2\pi}{3}\right) \\ i_T &= \sqrt{2} \cdot I_{fT} \sin\left(\omega \cdot t + \varphi - \frac{4\pi}{3}\right) \end{aligned} \quad (10)$$

$$M_{ik} = \frac{\mu_0}{2\pi} \cdot l \cdot \ln\left(\frac{D_{cp}}{d_{ik}}\right), \quad (12)$$

where $i \in (R, S, T)$, respectively $k \in (r, s, t)$.

$$\begin{aligned} U_r &= -j \cdot \omega \cdot (i_R \cdot M_{Rr} + i_S \cdot M_{Sr} + i_T \cdot M_{Tr}) \\ U_s &= -j \cdot \omega \cdot (i_R \cdot M_{Rs} + i_S \cdot M_{Ss} + i_T \cdot M_{Ts}) \\ U_t &= -j \cdot \omega \cdot (i_R \cdot M_{Rt} + i_S \cdot M_{St} + i_T \cdot M_{Tt}) \end{aligned} \quad (13)$$

a) The average distance between the conductors and the return path through earth is obtained taking into account the earth resistivity, using the following relation:

$$D_{CP} = 550 \sqrt{\frac{\rho}{f}}, \quad (1)$$

where ρ - the earth resistivity and f - line voltage frequency.

Example of inductive coupling effect

Table 5. Comparison between the measured and calculated voltages induced by magnetic coupling,.

Overhead electric line	Circ. 1 [km]	Circ. 2 [km]	Active circuit voltage	Active circuit current	Magnetically induced voltage measured in the passivated circuit			Magnetically induced voltage calculated in the passivated circuit		
			U [kV]	I [A]	U _R [V]	U _S [V]	U _T [V]	U _r [V]	U _s [V]	U _t [V]
Mintia-Hasdat	49.876	25.455	237.5	265.75	320	21	120	317.193	21.014	120.452
Peșt.-Oțelarie. max. charge	25.455	11.249	236.9	156.9	68	10.2	39	67.888	10.365	39.073
Pestis-Otelarie	25.455	11.249	236.9	74.826	34	10.2	18.5	33.415	7.18	18.808
Baru M-Hasdat	16.688	43.897	225	181.68	120	26	122	120.222	25.88	120.363
Otelarie-Hasdat	25.455	7.422	236.8	92.542	11	4.3	13.2	11.279	4.28	13.584
PdF-Resita	116.55	116.55	228	440	1400	400	1440	1401	404.28	1444
			225	480						
			232	460						
Mintia-Pestis	18.675	18.675	237	71.77	63.6	13	42	62.844	13.182	42.529
Resita - Iaz	30.730	30.730	226.5	14.54	20.8	3.6	17	20.862	3.671	17.091
Resita -Timisoara	72.867	72.867	234	497.03	1020	400	940	1021	357.93	941.014
Timisoara-Sacalaz	53.719	24.620	230	212	180	71	1260	181.052	71.05	1261
			235	237						
			225	218						
Sacalaz-Arad	53.719	55.173	230	212	310	80	270	311.44	81.009	270.174
			235	237						
			225	218						

(Suaranu 2009)

Exercises



Exercise 1: Capacitive coupling matrix

- ◆ Two conductors of 100 mm diameter, conductor 1 is 5m above ground, conductor 2 is 1 m higher and directly above conductor 1.

Determine the capacitance matrix.

Note: The vacuum permittivity is $\epsilon_0 = 8.8542 \cdot 10^{-12} \text{ Fm}^{-1}$

Exercise 2: Railway transmission line and pasture fence

- ♦ There is a pasture fence (f), that is capacitively influenced by a two-phase (a,b) 110 kV railway transmission line of $f = 16,7$ Hz.

The capacitance matrix of the system is given by

$$c'_{abf} = \begin{bmatrix} 8,910 & -1,807 & -0,048 \\ -1,807 & 8,135 & -0,091 \\ -0,048 & -0,091 & 11,508 \end{bmatrix} \frac{nF}{km}$$

- Calculate the touch voltage of the fence
- Calculate the touch current of the fence
- What is the maximum length of the fence in parallel to the line, so that a touch current of 2 mA is not exceeded?

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